MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, CAST, 90-DEGREE E-PLANE BEND, FULL CORRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.

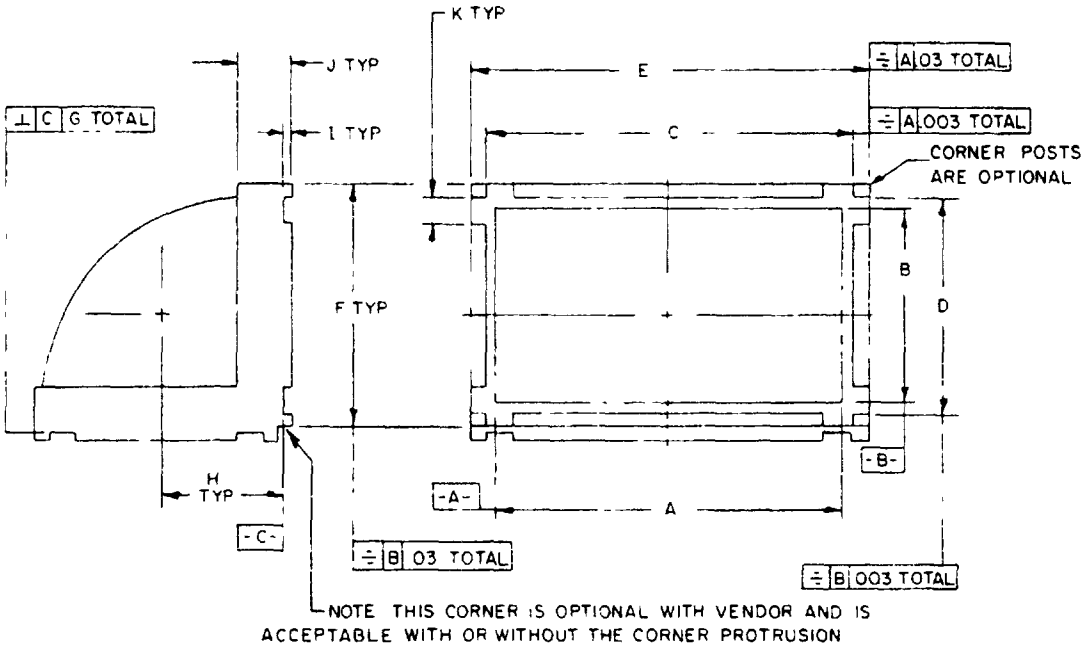


FIGURE 1 Dimensions for configuration 1, for dash numbers 001, 002, 003, 006, 007, 008, 009, 012, 013, 016, and 017

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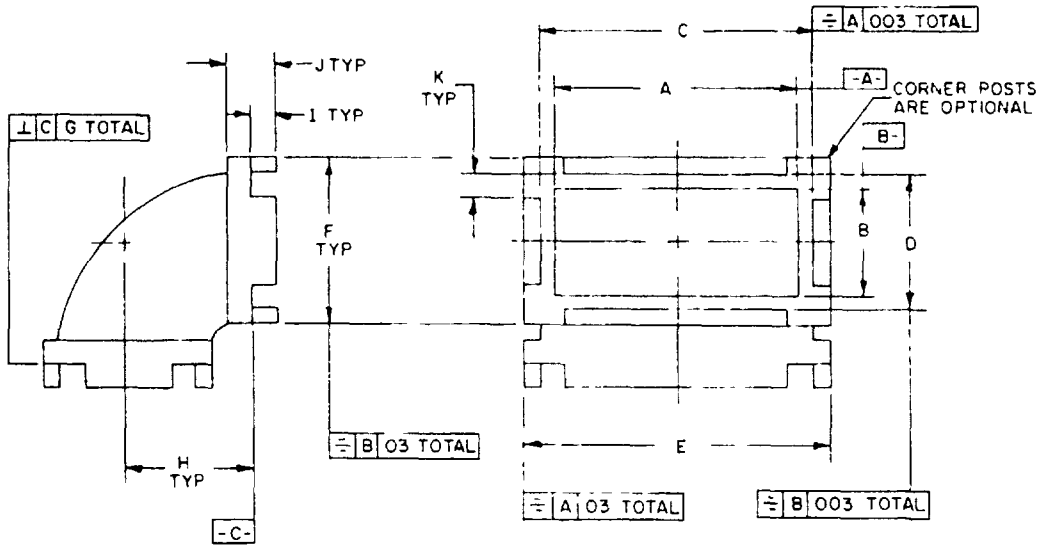


FIGURE 2 Dimensions for configuration 2, for dash numbers 010 and 011

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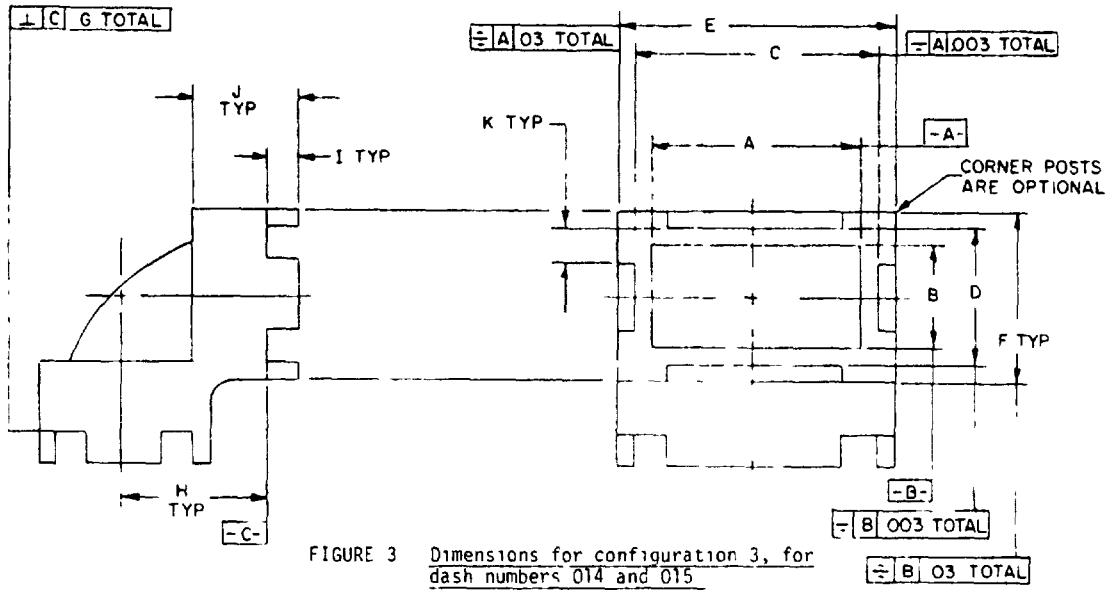


FIGURE 3 Dimensions for configuration 3, for dash numbers 014 and 015

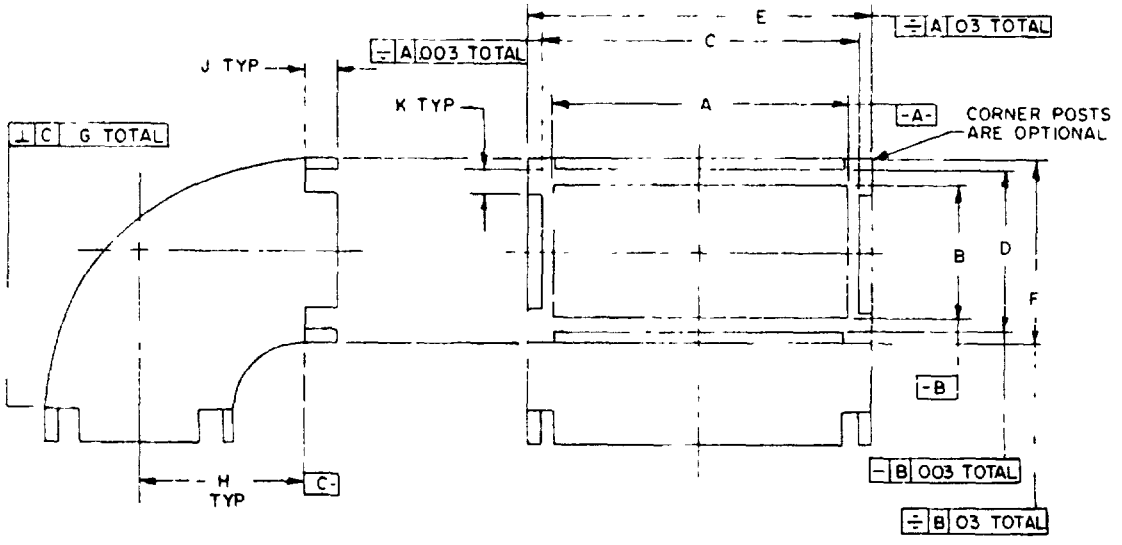


FIGURE 4 Dimensions for configuration 4, for dash numbers 004 and 005

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TABLE I Dash number, electrical and physical characteristics, and material.

Dash No.	Frequency range TE_{10} mode GHz	VSWP max	Peak power kW max	Average power kW max	Operating temperature range	Pressurization lb_f/in^2 gage	Material
001	3.95 - 5.85	1.05	250	10	-55° to +300°C	45	Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
002	7.05 - 10.0		1/	1/	0° to 60°C		Beryllium copper, alloy X6 per QQ-A-390
003	7.05 - 10.0		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
004	7.05 - 10.0		1/	1/	0° to 60°C		Beryllium copper, alloy X6 per QQ-A-390
005	7.05 - 10.0		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
006	8.2 - 12.4		1/	1/	0° to 60°C		Beryllium copper, alloy X6 per QQ-A-390
007	8.2 - 12.4		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
008	8.2 - 12.4		1/	1/	0° to 60°C		Beryllium copper, alloy X6 per QQ-A-390
009	8.2 - 12.4		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
010	8.2 - 12.4		1/	1/	0° to 60°C		Beryllium copper, alloy X6 per QQ-A-390
011	8.2 - 12.4		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
012	12.4 - 18.0		1/	1/	0° to 60°C		Beryllium copper, alloy X6 per QQ-A-390
013	12.4 - 18.0		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
014	12.4 - 18.0		1/	1/	0° to +60°C		Beryllium copper, alloy X6 per QQ-A-390
015	12.4 - 18.0		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371
016	26.5 - 40.0		1/	1/	0° to +60°C		Beryllium copper, alloy X6 per QQ-A-390
017	26.5 - 40.0		250	10	-55° to +300°C		Aluminum alloy, D712.0, temper F, per QQ-A-601 or C712.1 per QQ-A-371

1/ The assemblies shall have a power capability of 80 percent of the mating waveguide average power at sea level pressure.

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TABLE II Dimensions, figures, and mating waveguides

Dash No	Dimensions 1/ 2/ A	B	C	D	E	F	G TOL	H	I	J	K	FIGURE	Mating Waveguide M85/
001	1 875 ± 005	872 ± 005	2 005 + 005 - 000	1 005 + 005 - 000	2 18 ± 03	1 18 + 03	± 009	593 ± 010	06 ± 01	18 ± 01	180 ± 006	1	1-053 1-054 1-174
002	1 122 ± 004	497 ± 004	1 252 + 003 - 000	627 + 003 - 000	1 36 ± 03	74 ± 03	± 005				.112 ± 004	1	1-069 1-073
003	1 122 ± 004	497 ± 004	1.252 + 003 000	627 + 003 - 000	1 36 ± 03	74 ± 03					112 ± 004	1	1-071 1-072 1-177
004	1 122 ± 004	497 ± 004	1 252 + 003 000	627 + 003 - 000	1 343 ± 010	718 ± 010					112 ± 004	4	1-069 1-073
005	1 122 + 004	497 ± 004	1 252 + 003 - 000	627 + 003 - 000	1 343 ± 010	718 ± 010					.112 ± 004	4	1-071 1-072 1-177
006	900 ± 003	400 ± 003	1 002 + 003 - 000	502 + 003 - 000	1 15 ± 03	65 + 03	± 004	453 ± 010			090 ± 003	1	1-075 1-079
007	900 ± 003	400 +.003	1 002 + 003 - 000	502 + 003 - 000	1 15 + 03	65 ± 03	± 004	453 ± 010			090 ± 003	1	1-077 1-078 1-178
008	900 ± 003	400 ± 003	1 003 + 003 - 000	503 + 003 - 000	1 125 + 010	625 + 010	± 004	328 ± 010			090 ±.003	1	1-075 1-079
009	900 ± 003	400 ± 003	1 003 + 003 000	503 + 003 - 000	1 125 ± 010	625 + 010	± 004	328 ± 010			090 + 003	1	1-077 1-078 1-178

See footnotes on page 6

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
000	00	03	8	453	11 51	718	18 24	1 125	28 58
003	08	06	1 5	497	12 62	74	18 8	1 15	29 2
004	10	090	2 29	502	12 75	872	22 15	1 18	30 0
005	13	112	2 84	503	12 78	900	22 86	1 252	31 80
006	15	180	4 57	593	15 06	1 002	25 45	1 343	34 11
009	23	18	4 6	625	15 88	1 003	25 48	1 36	34 5
010	25	328	8 33	627	15 93	1 005	25 53	1 875	47 62
01	3	400	10 16	65	16 5	1 122	28 50	2.005	50.93
								2 18	55 37

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TABLE II Dimensions, figures, and mating waveguides - Continued.

Dimensions 1/ 2/													Mating Waveguide M85
Dash No	A	B	C	D	E	F	G TOL	H	I	J	K	FIGURE	
010	900 ± 003	400 ± 003	1 003 + 003 - 000	503 + 003 - 000	1 125 ± 010	625 ± 010	± 004	453 ±.010	06 ±.01	18 ±.01	090 ±.003	2	1-075 1-079
011	900 ± 003	400 ± 003	1 003 + 003 - 000	503 + 003 - 000	1.125 ± 010	625 ± 010	± 004	453 ± 010	06 ± 01	18 ± 01	090 ± 003	2	1-077 1-078 1-178
012	622 ± 002	311 ± 002	704 + 003 - 000	393 + 003 - 000	80 ± 03	58 ± 03	+ 003	406 ± 010	06 + 01	18 ± 01	063 ± 002	1	1-087 1-089
013	622 ± 002	311 ± 002	.704 + 003 -.000	393 + 003 -.000	80 ± 03	58 ± 03	± 003	406 ± 010	06 ± 01	.18 ± 01	.063 ± 002	1	1-090 1-091 1-180
014	622 + 002	311 ± 002	705 + 003 - 000	394 + 003 - 000	812 + 010	500 ± 010	± 003	406 ± 010	06 ± 01	18 ± 01	.063 ±.002	3	1-087 1-089
015	622 ± 002	311 ± 002	705 + 003 - 000	394 + 003 - 000	812 ± 010	500 ± 010	± 003	406 ± 010	.06 ±.01	.18 ± 01	063 ± 002	3	1-090 1-091 1-180
016	280 ± 0015	140 ± 0015	362 + 002 - 000	222 + 002 - 000	437 ± 010	296 ± 010	± 002	140 ± 010	04 ± 01	12 ± 01	.033 ± 001	1	3-007 3-008
017	280 ± 0015	140 + 0015	362 + 002 - 000	222 + 002 - 000	437 + 010	296 ± 010	+ 002	140 ± 010	04 ± 01	12 ± 01	033 + 001	1	3-009

NOTES
1/ Dimensions are in inches
2/ Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
000	00	033	84	222	5 64	406	10 31	704	17 88
0015	038	04	1 0	280	7 11	437	11 10	705	17 91
002	05	063	1 60	296	7 52	453	11 51	80	20 3
003	.08	06	1 5	311	7 90	500	12 70	812	20.62
004	10	090	2 29	362	9 19	503	12 78	900	22 86
010	25	12	3 0	393	9 98	58	14 7	1 003	25 48
01	3	140	3 56	394	10 01	622	15 80	1 125	28 58
03	8	18	4 6	400	10 16	625	15 88		

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REQUIREMENTS

Physical characteristics
Dimensions and configuration See figures 1, 2, 3, and 4, and tables I and II.
Material See table I.
Surfaces (sandblasting after machining permissible)
Internal $63\sqrt{}$ microinches per ANSI B46.1.
External $125\sqrt{}$ microinches per ANSI B46.1

Fungus resistance Materials nutrient to fungus shall not be used

Pressurization See table I

Penetrant inspection No indication of any cracks at any location when tested per MIL-I-6866, type I, method B.

Electrical characteristics See table I

Part number M3970/22-(dash number from table I).

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI, SG
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AS
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5985-0881-2)

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